Ex.No: 09	PREDICTION USING LINEAR REGRESSION
Date:	

AIM:

To perform the time series analyze in POWER BI by using python script and visualize the charts.

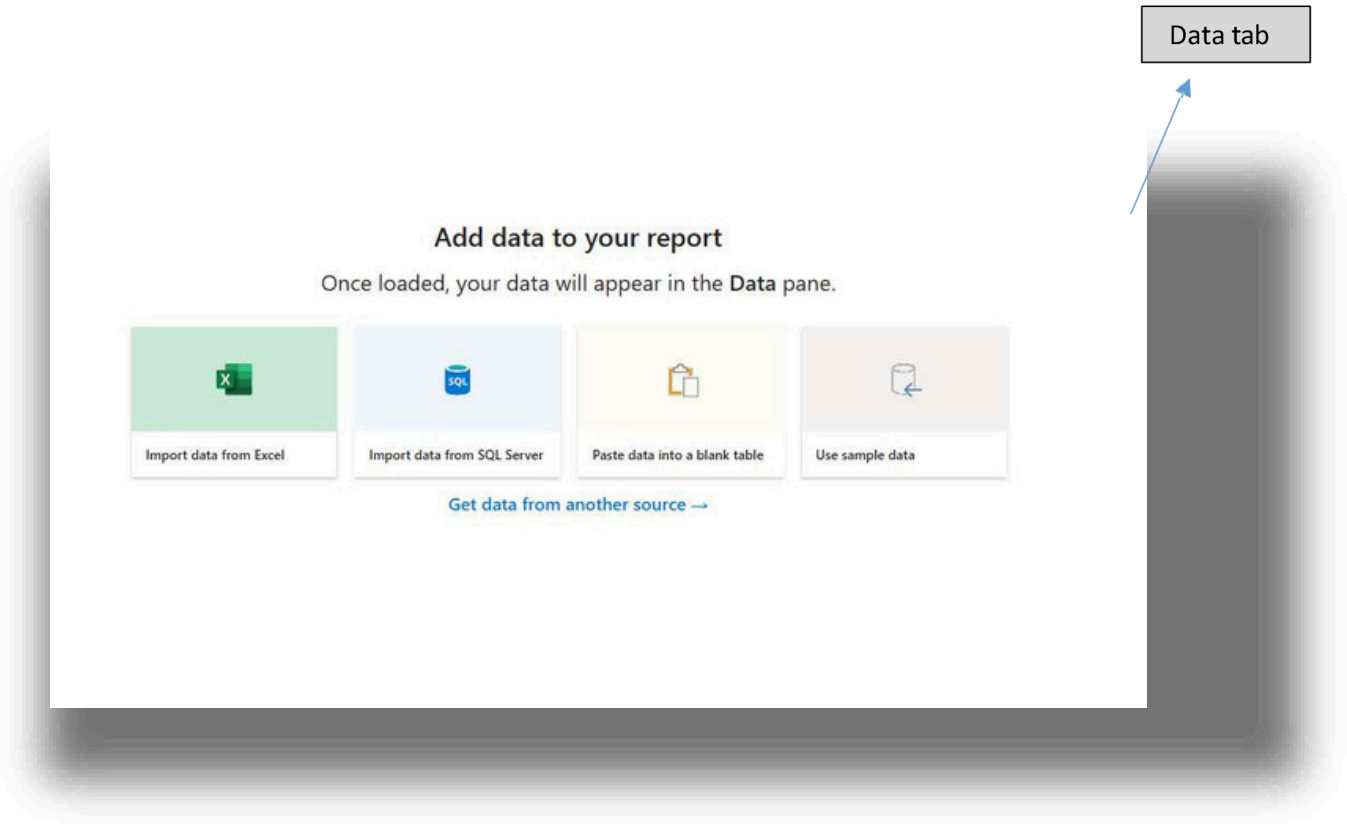
IMPLEMENTATION:

1. Install and Open Power BI

1. **Download & Install Power BI**
   - Visit the official Power BI Download Page.
   - Download and install **Power BI Desktop**.
2. **Open Power BI Application**
   - Launch the Power BI Desktop application after installation.

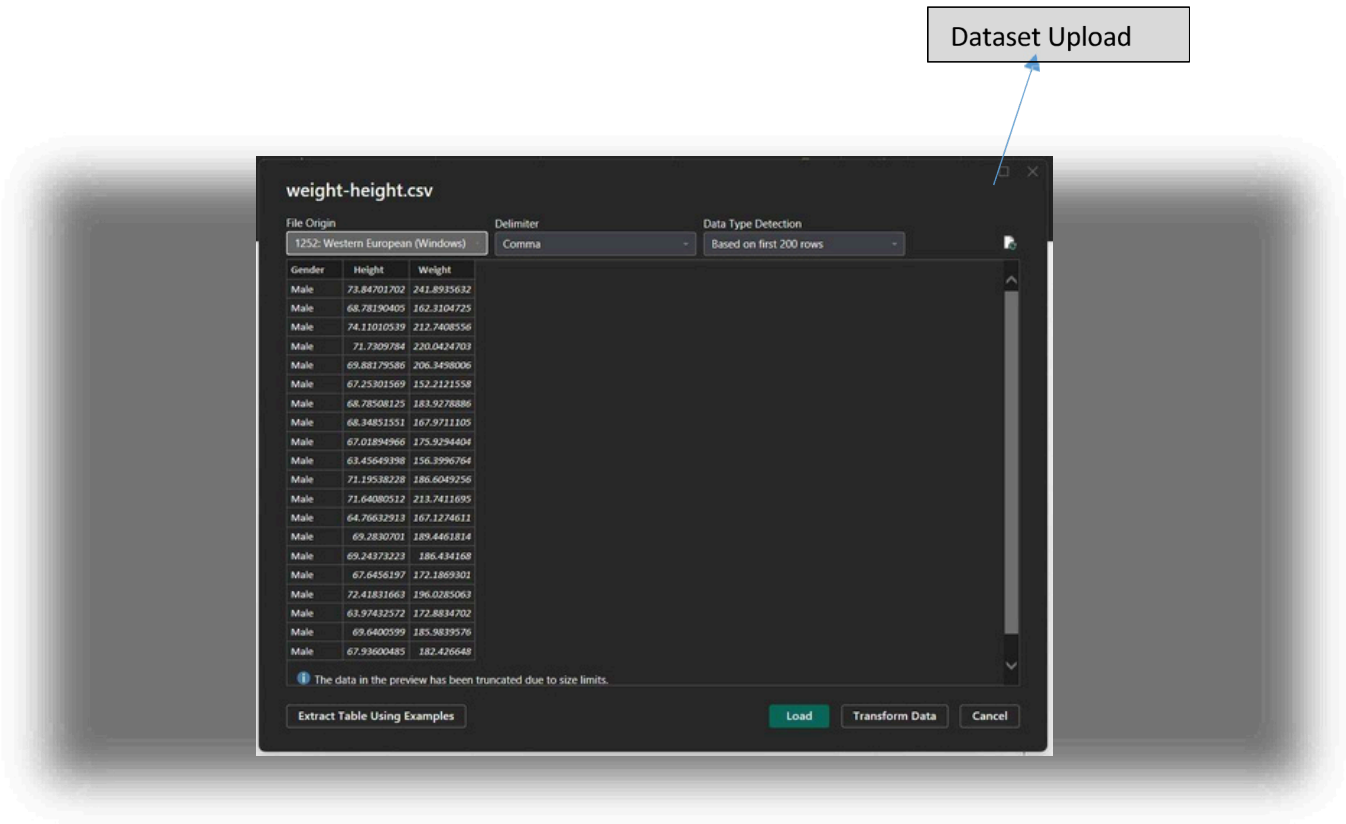
2. Load the Time Series Dataset

3. **Download the Dataset**
   - Ensure you have a **time series dataset** in a suitable format (CSV, TXT, etc.).
   - Example dataset: **"time_series_sales_dataset.csv"**.
4. **Import the Dataset into Power BI**
   - Open **Power BI Desktop**.
   - Click **Home > Get Data > Text/CSV**.
   - Select **"time_series_sales_dataset.csv"** and click **Load**.
   - Ensure that the **Date column** is formatted as **Date/Time**.
5. **Verify the Imported Data**
   - Click the **Data View** option (right panel) to preview the dataset.



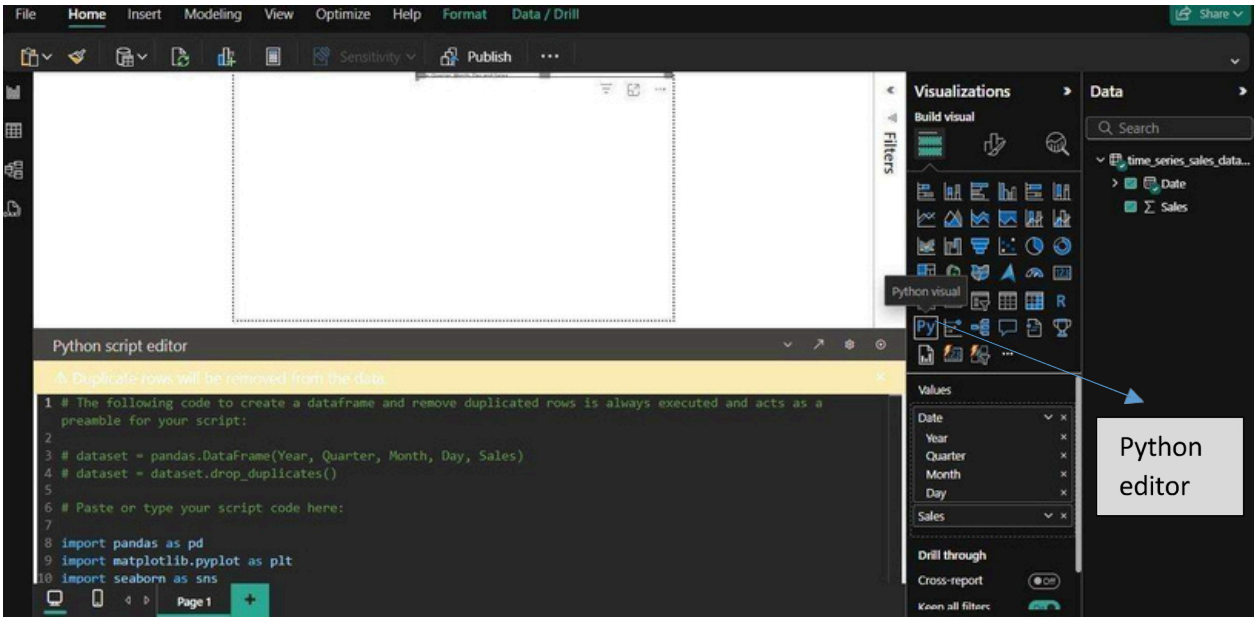
3. Visualize Time Series Data in Power BI

6. **Open the Visualization Pane**
   - Click the **Report View** tab.
   - Click the **"Visualization"** pane to see all available visualizations.
7. **Create a Time Series Line Chart (optional)**
   - From **Visualizations**, select the **Line Chart**.
   - **Drag Date** to the **X-axis**.
   - **Drag Sales** to the **Y-axis**.
   - Title the chart as **"Time Series Sales Analysis"**.



4. Run Python Script for Advanced Analysis

8. **Enable Python Scripting in Power BI**
   - Click **File > Options and Settings > Options**.
   - Under **Python scripting**, set the Python home directory (Anaconda/Miniconda).
9. **Add Python Script in Power BI**
   - Click **"Python Visual" (Py icon)** in the **Visualization Pane**.
   - Select the required **columns** from the dataset.

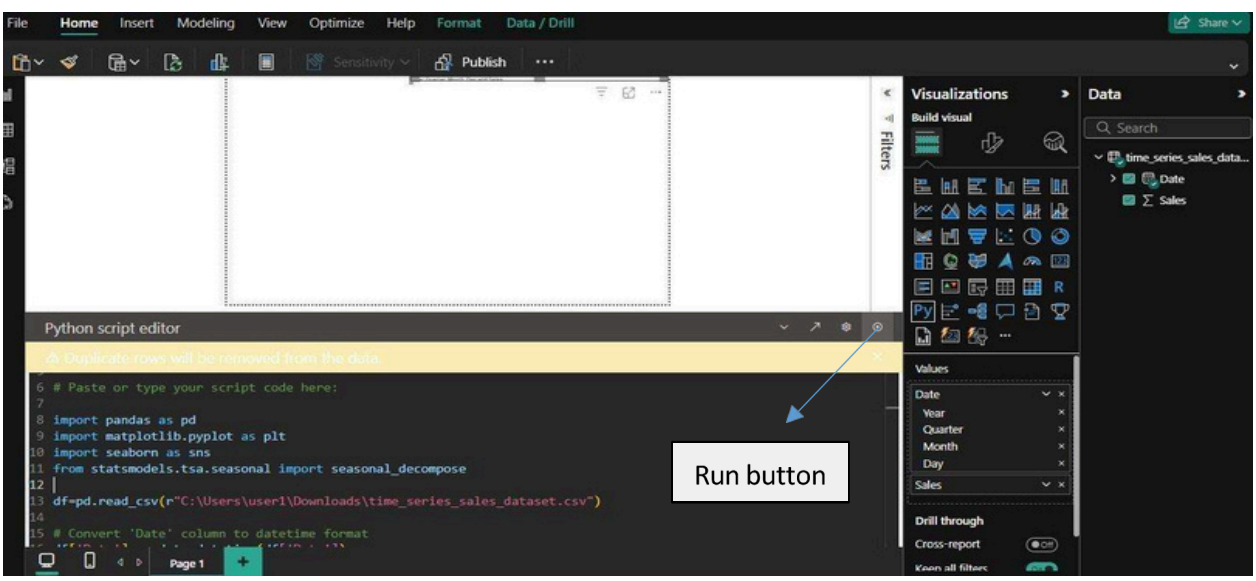


10. **Paste and Run the Python Script**

- Open the **Python Script Editor** in Power BI.
- Paste the following **Python script**:

11. **Run the Python Script**

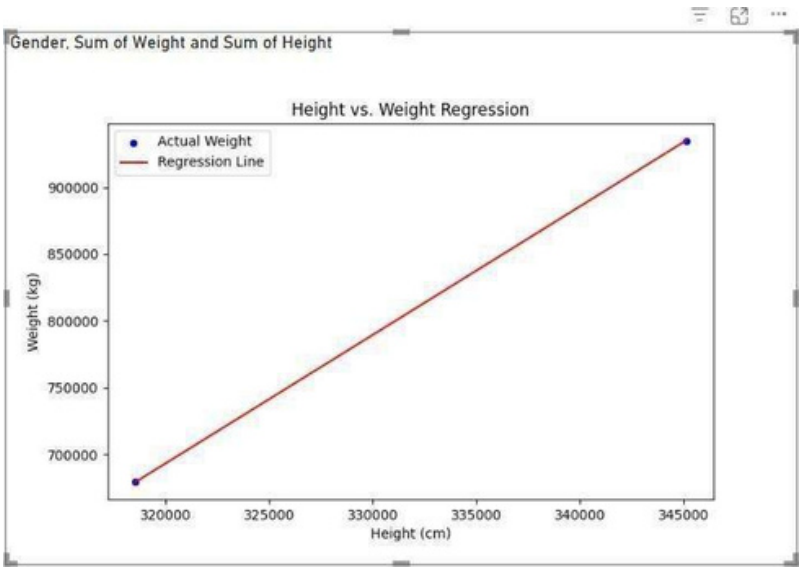
- Click the **Run button** on the top of the Python script editor.
- Wait for execution to complete.



CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression

# Load data from Power BI
dataset = dataset  # Power BI automatically passes the dataset

# Define independent (X) and dependent (Y) variables
X = dataset[['Height']]  # Predictor
y = dataset['Weight']  # Response Variable

# Create and train the Linear Regression model
model = LinearRegression()
model.fit(X, y)

# Predict weight using height
dataset['Predicted_Weight'] = model.predict(X)

# Plot the Regression Line
plt.figure(figsize=(8,5))
sns.scatterplot(x=X['Height'], y=y, color='blue', label='Actual Weight')
sns.lineplot(x=X['Height'], y=dataset['Predicted_Weight'], color='red', label='Regression
Line')
plt.xlabel("Height (cm)")
plt.ylabel("Weight (kg)")
plt.title("Height vs. Weight Regression")
plt.legend()
plt.show()
```

OUTPUT:



RESULT:

The program was executed successfully.